StreamPixel | Ridith Shetty | PERFBOARD BUILD — NO PCB ORDER

GPIO labels are logical connections, NOT a physical 30-pin header drawing. Verify the actual board labels.

Panel OUT -> next panel IN. Separate fused power to every panel. Do not carry panel current through perfboard.

U1

ESP32 30-pin dev board — GPIO LABELS

U2

SN74HCT245N (A-to-B)

U3

SN74HCT245N (A-to-B)

J1

HUB75 PANEL 1 IN — VERIFY PINOUT

J4

PIN8: select GND OR E after verification

U4

MAX9814 — follow printed pin labels

C1

100nF at U2 pins20/10

C2

100nF at U3 pins20/10

C3

470uF 25V — polarity matters

R1

10k OE blank at reset

5V buffer outputs NEVER connect to ESP32 inputs. U2/U3 must be HCT parts, not HC-only replacements.

MAX9814 powered at 3.3V; verify actual output range before connecting GPIO34 (ADC1).

J4 stays open until panel pin8 is identified. Never bridge both positions or ground a driven E output.

J2

5V DC from protected supply — NO MAINS HERE

F1

1A controller

F2

5A candidate — verify branch load

J5

Panel 1 DC power lead

F3

5A candidate — verify branch load

J6

Panel 2 DC power lead

F4

5A candidate — verify branch load

J7

Panel 3 DC power lead

Supply case and terminals require suitable protection, earth and strain relief. This sheet covers LOW-VOLTAGE DC wiring only.

Disconnect external supply and display harness for USB programming unless dev-board power isolation is documented.

NO CUSTOM PCB REQUIRED. Logical module pin labels; verify actual hardware.

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